

LEAD SCORE CASE STUDY

LOGISTIC REGRESSION MODEL FOR LEAD CONVERSION

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OBJECTIVE

- Identify significant features influencing conversion.
- Develop and validate a logistic regression model.
- Evaluate performance with respect to accuracy, sensitivity and specificity.



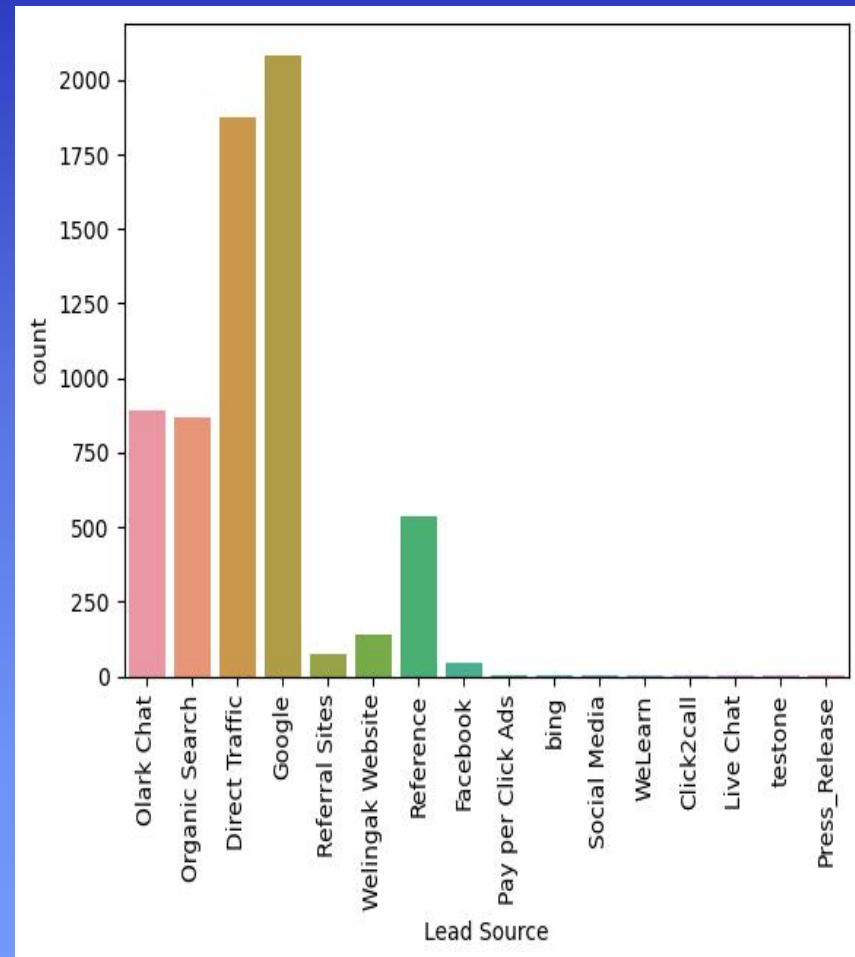
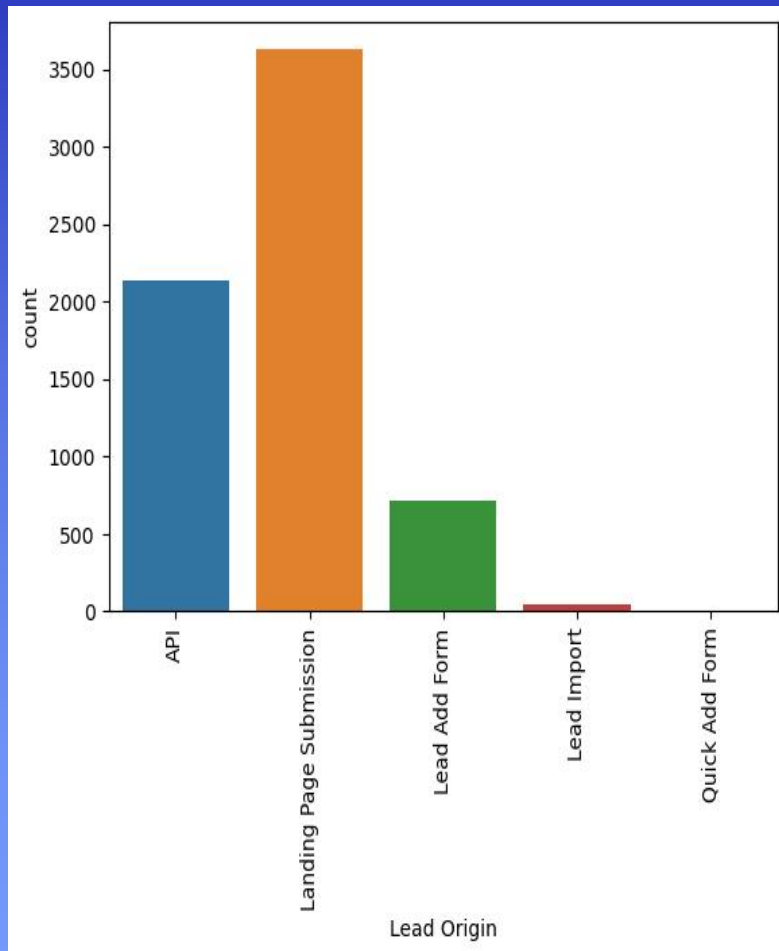
OVERVIEW

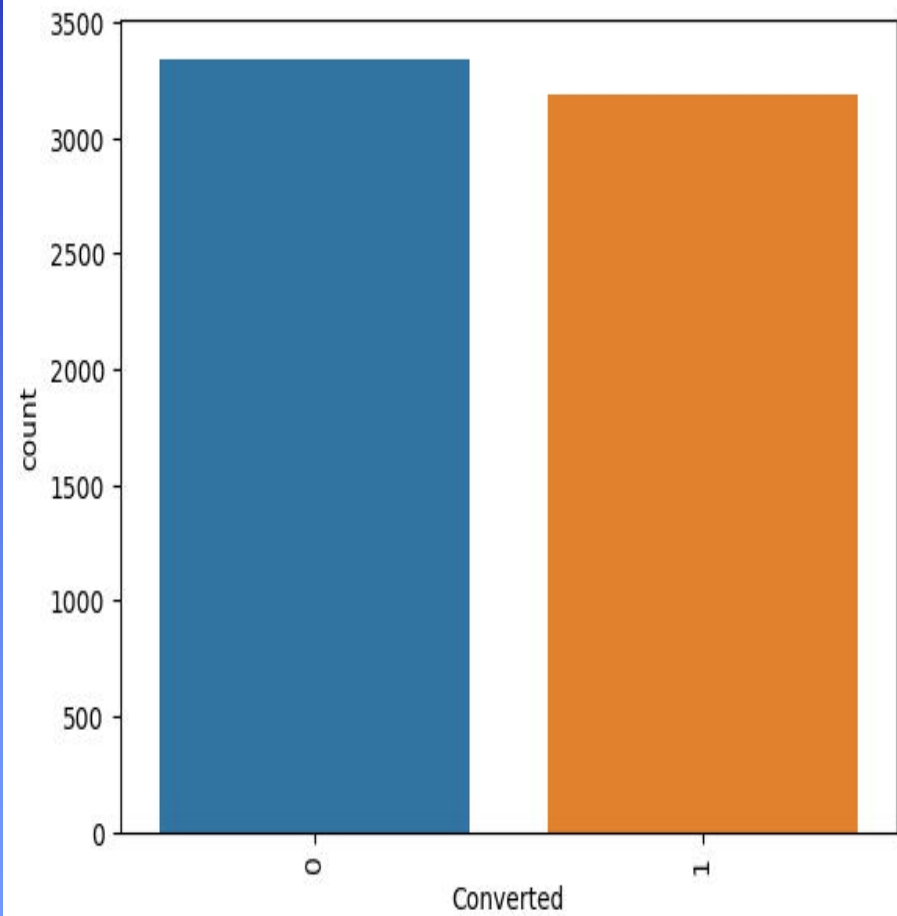
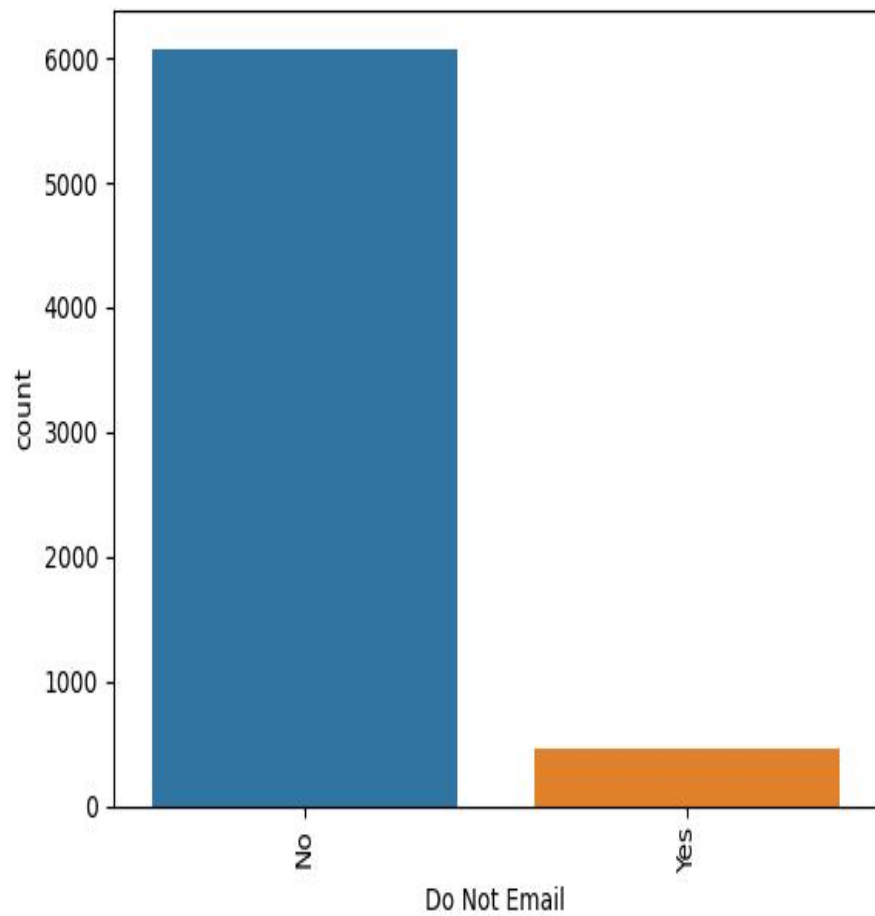
- **Keys Features:-**

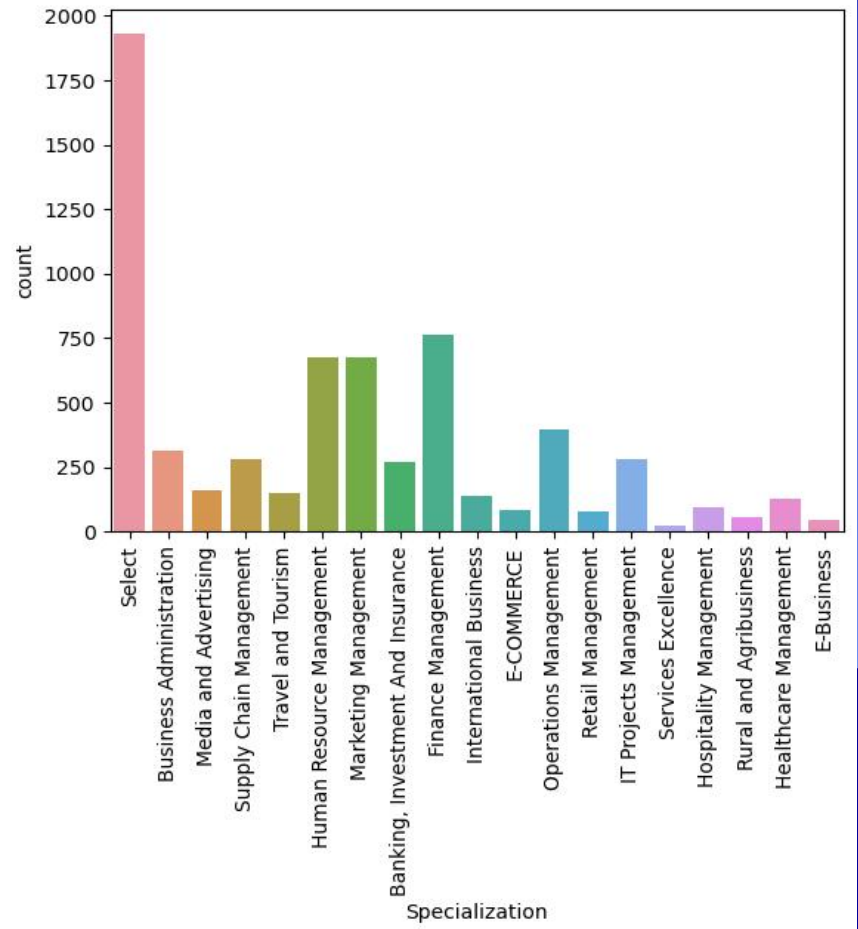
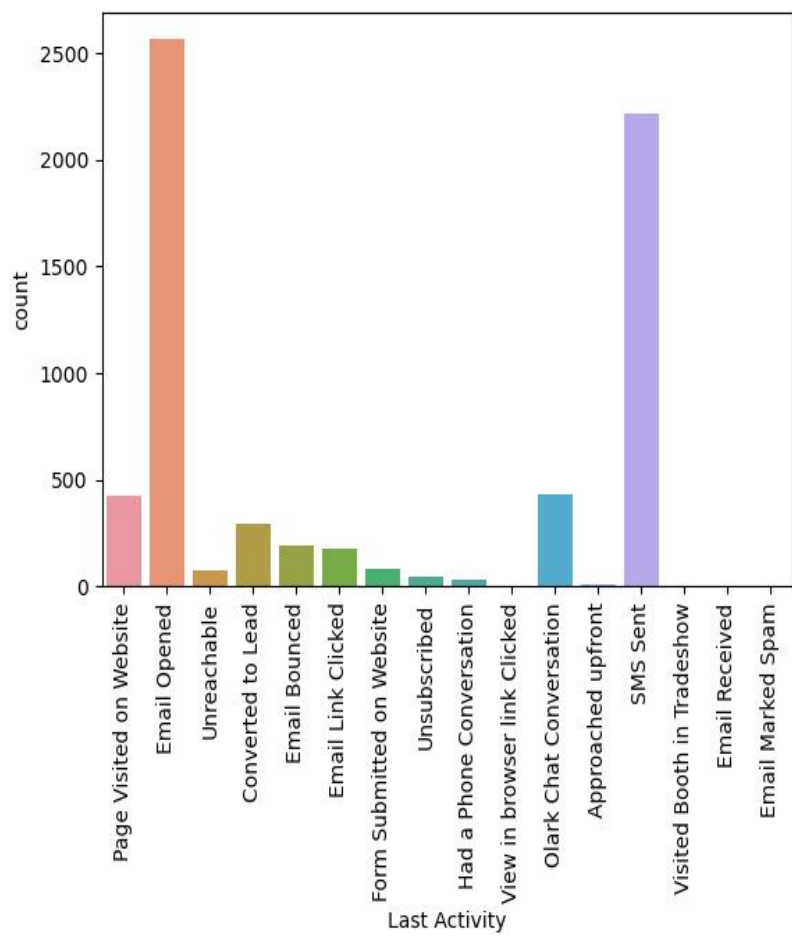
- Total Visits
- Total Time Spent On Website
- Lead Origin Lead Add Form
- Last Notable Activity Unreachable
- Lead Source Welingak Website

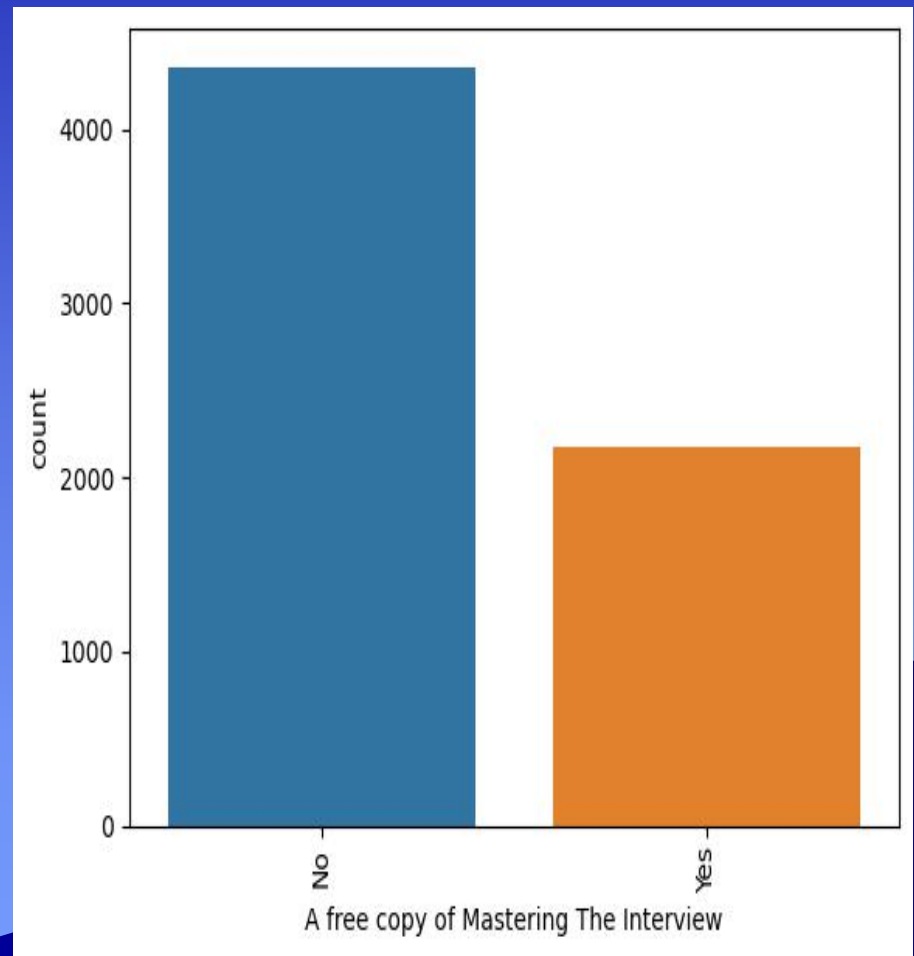
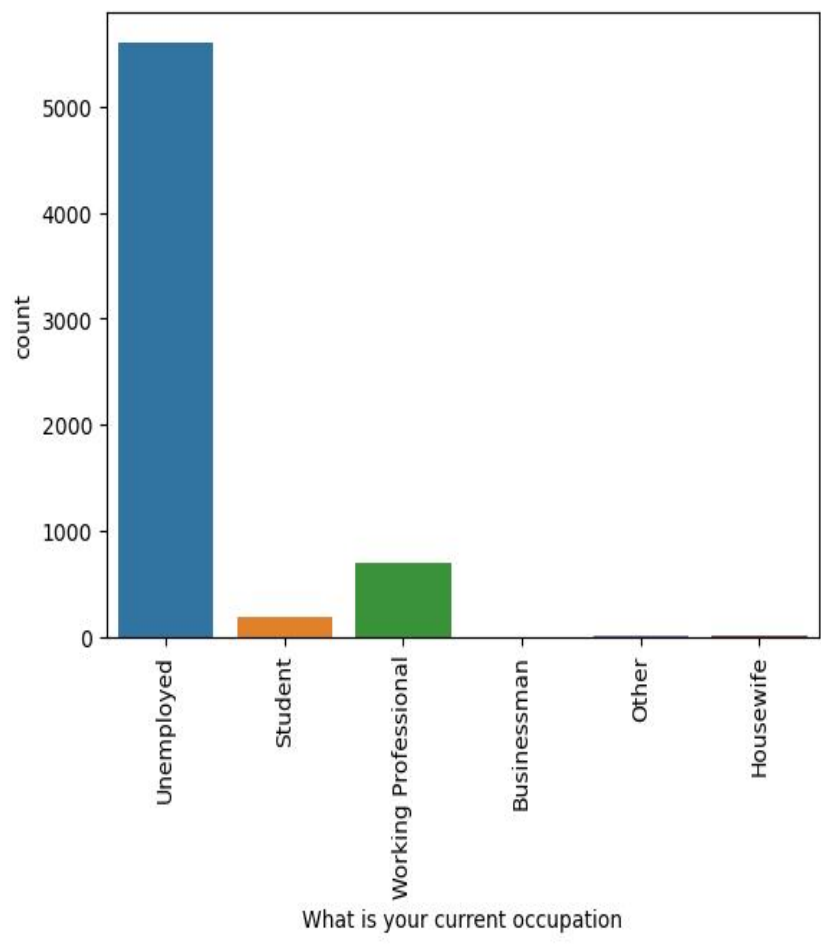


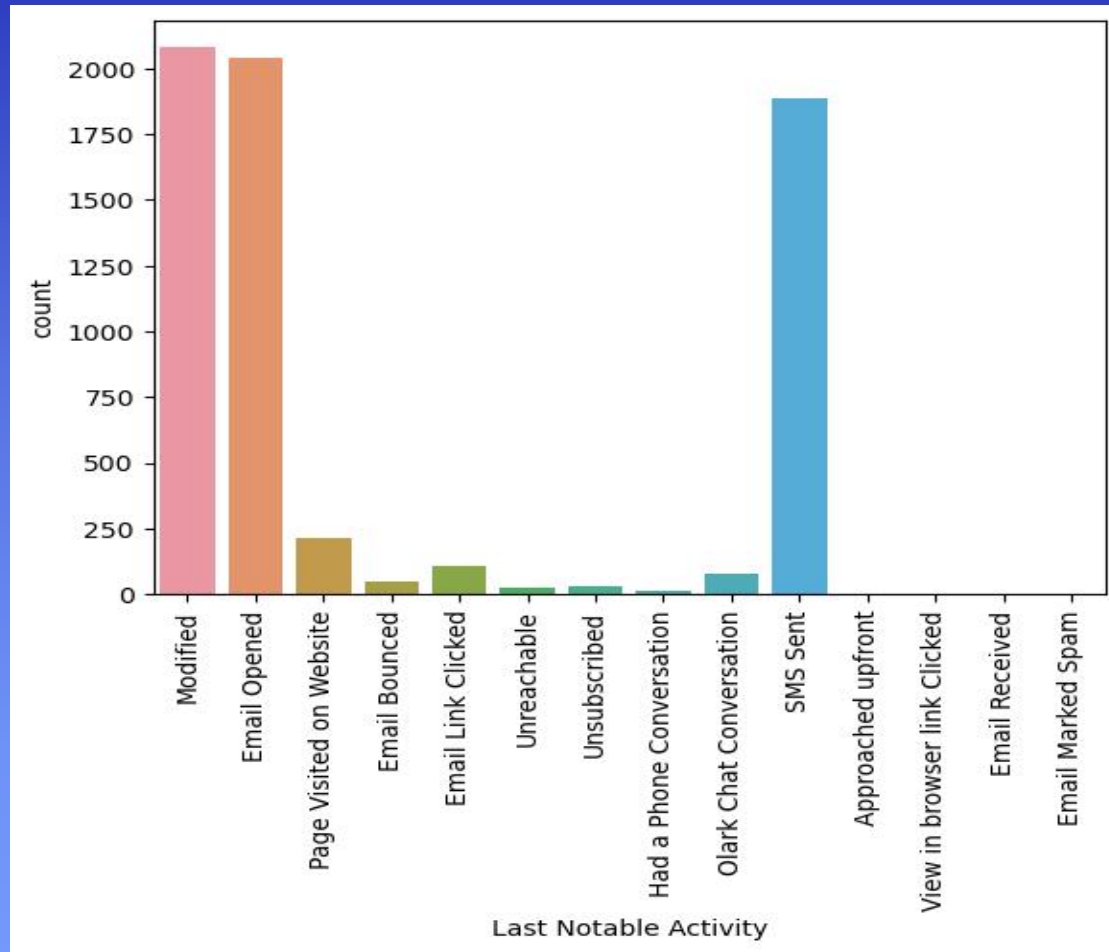
UNIVARIATE ANALYSIS (COUNT PLOT)



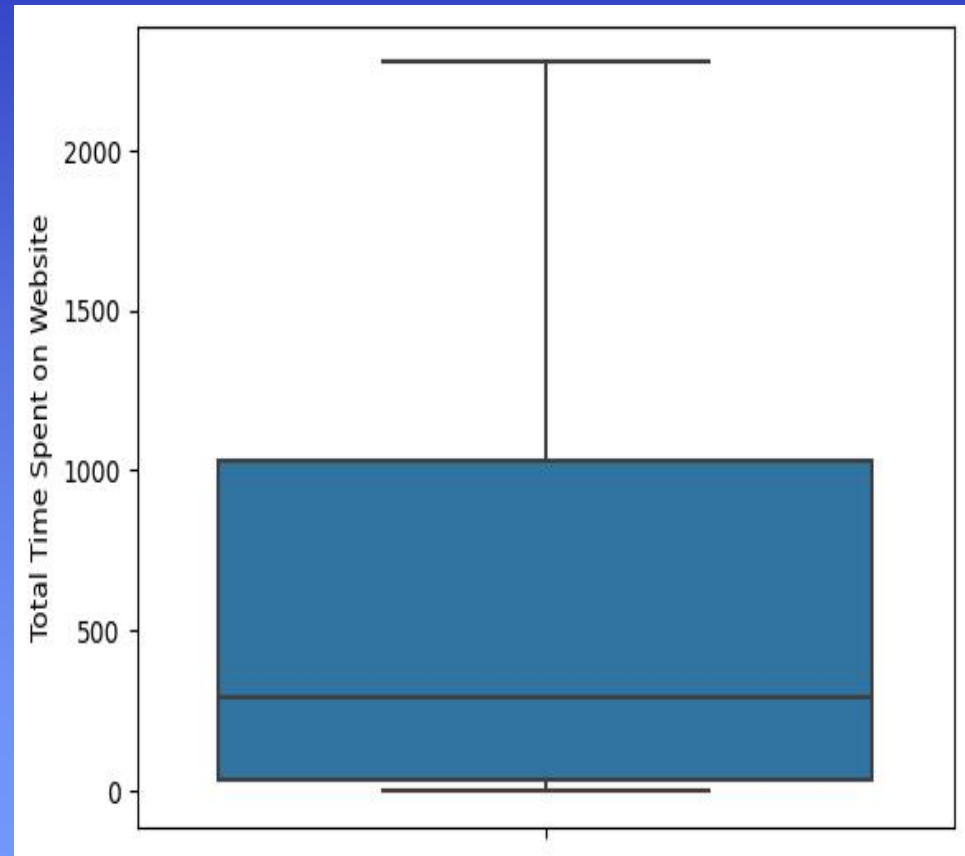
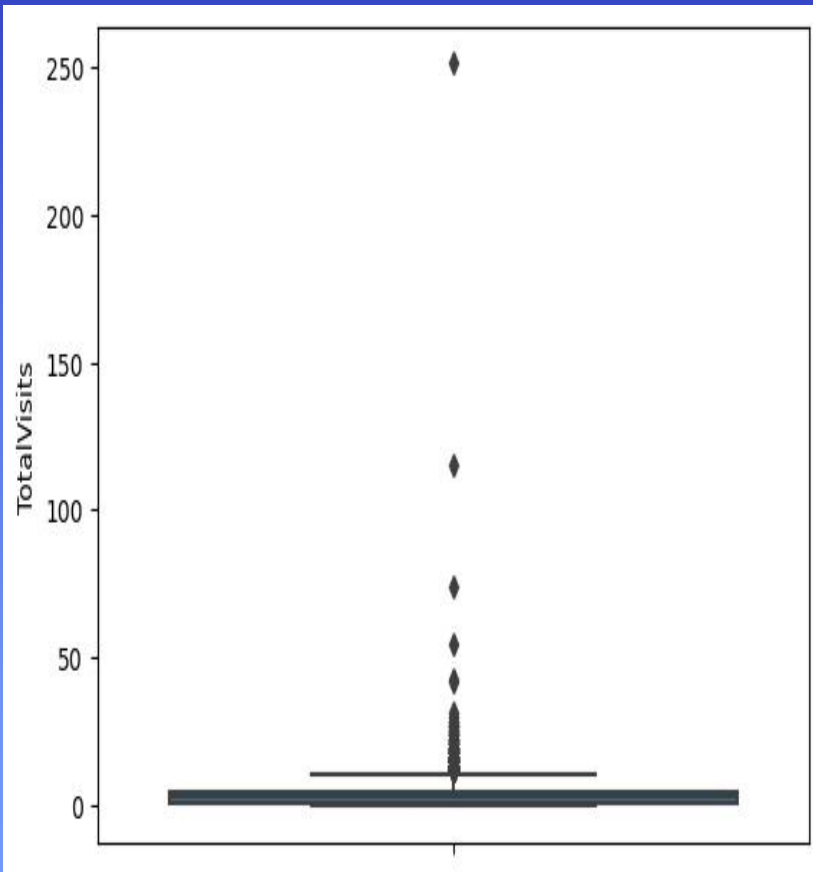


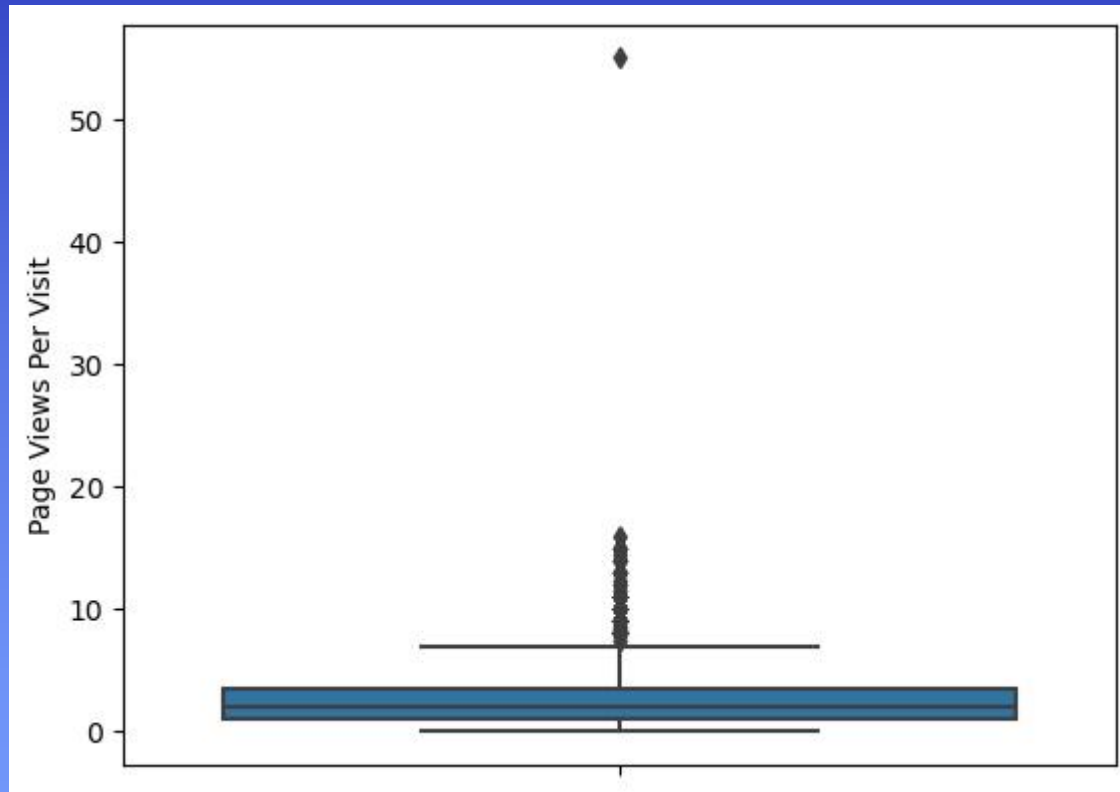




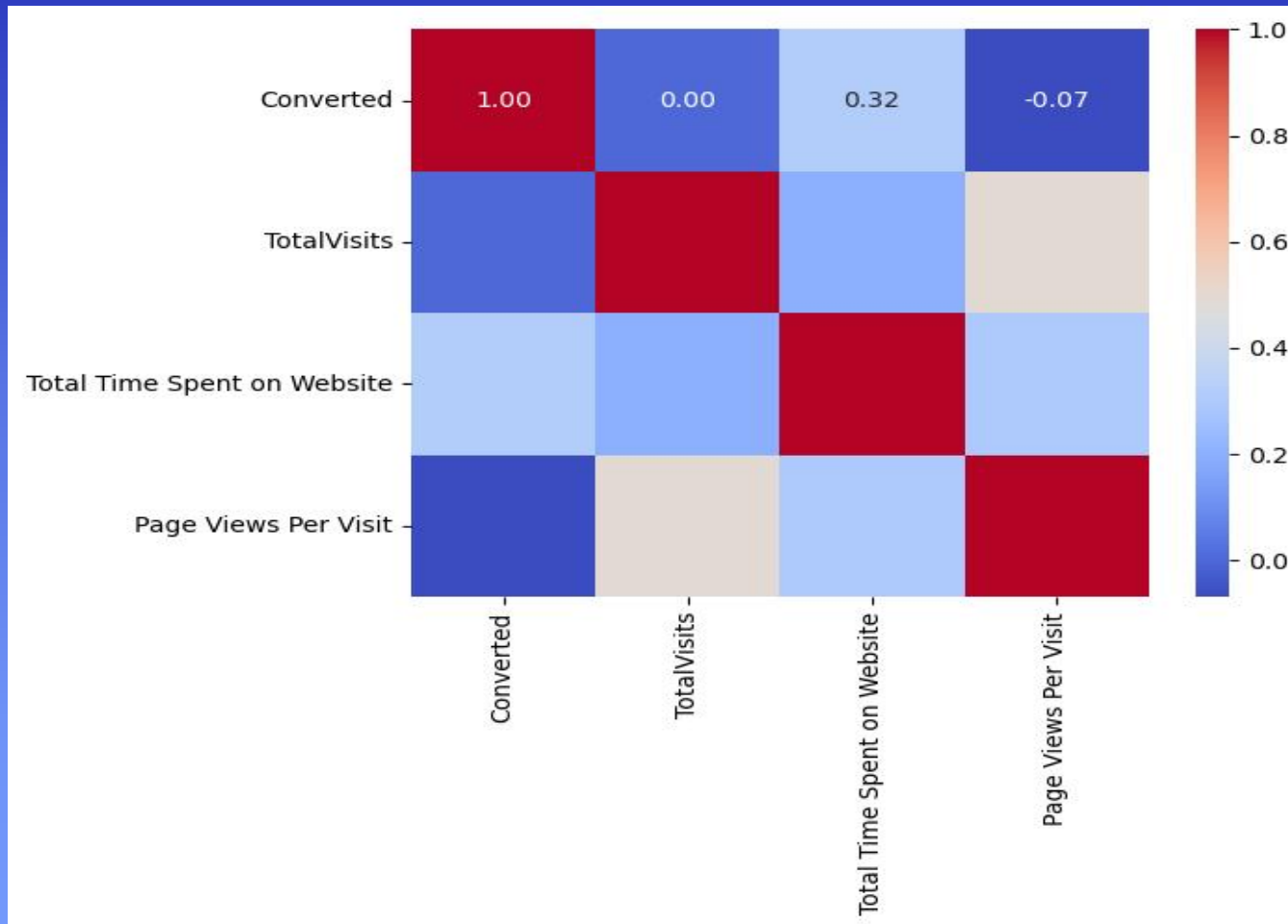


BIVARIATE ANALYSIS - BOXPLOT FOR OUTLIER ANALYSIS





MULTIVARIATE ANALYSIS - (HEATMAP)



LOGISTIC REGRESSION MODEL

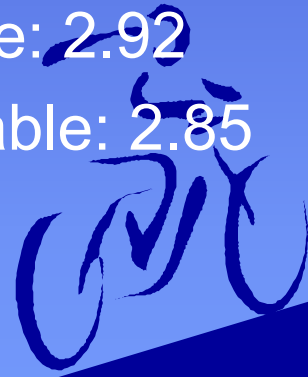
- Model Selection: Logistic Regression
- Target Variable: Converted

Variables & Coefficients:

- Total Visits: 10:48
- Total Time Spent On The Website: 4.39
- Lead Source - Welingak Website: 2.92
- Last Notable Activity - Unreachable: 2.85

Multicollinearity Check:

VIF < 5 for all variables



Feature Selection

- **Categorical Features:**

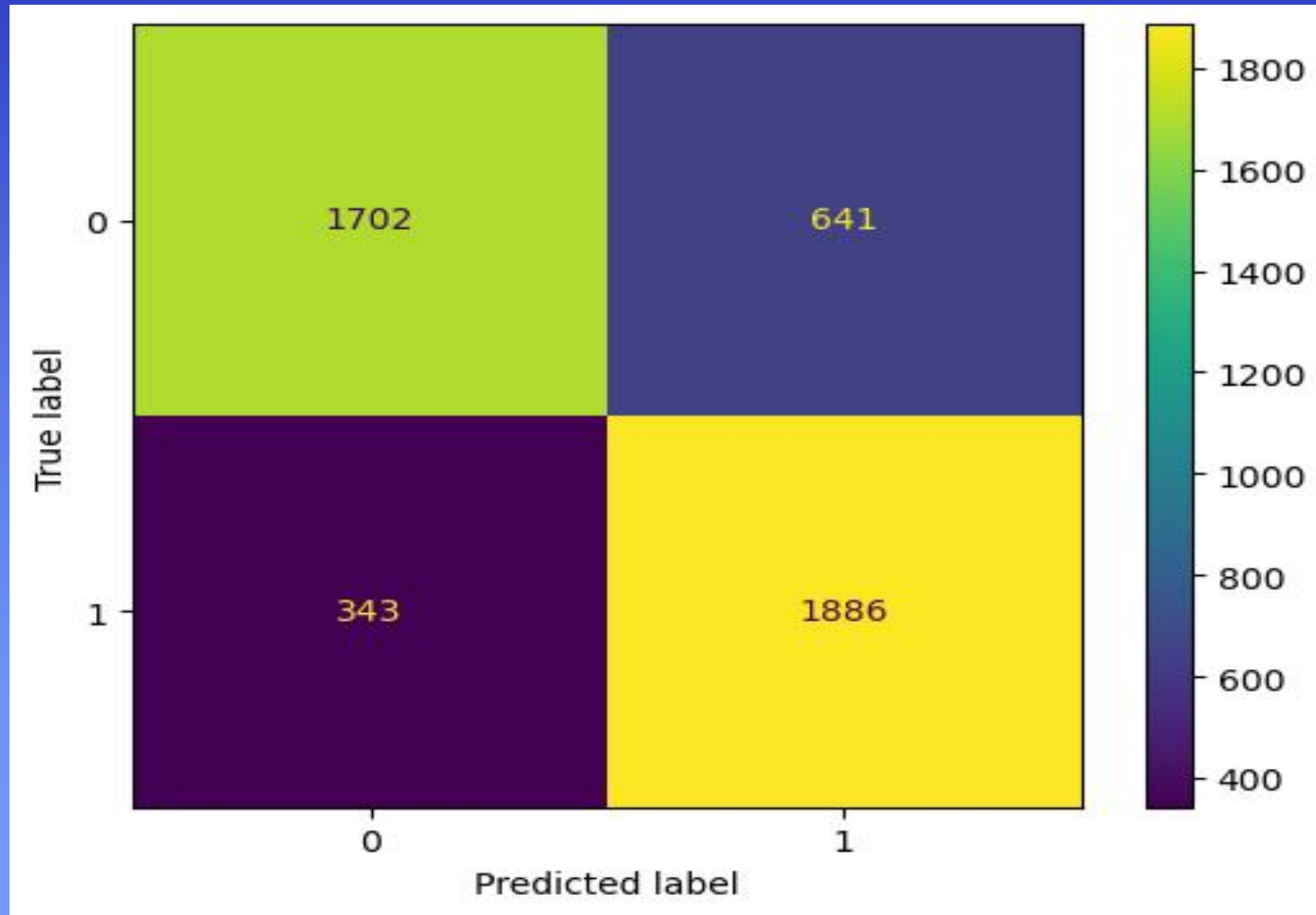
1. Lead Source - Welingak Website
2. Lead Origin - Lead Add Forms
3. Last Notable Activity - Unreachable

- **Numerical Features:**

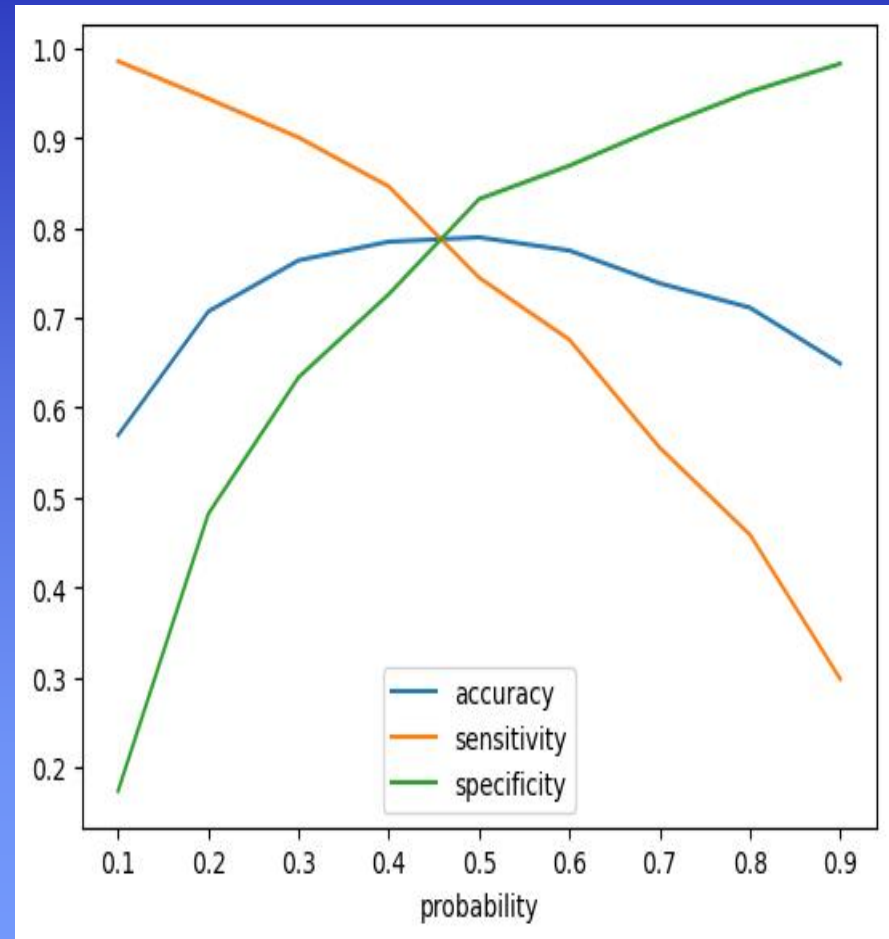
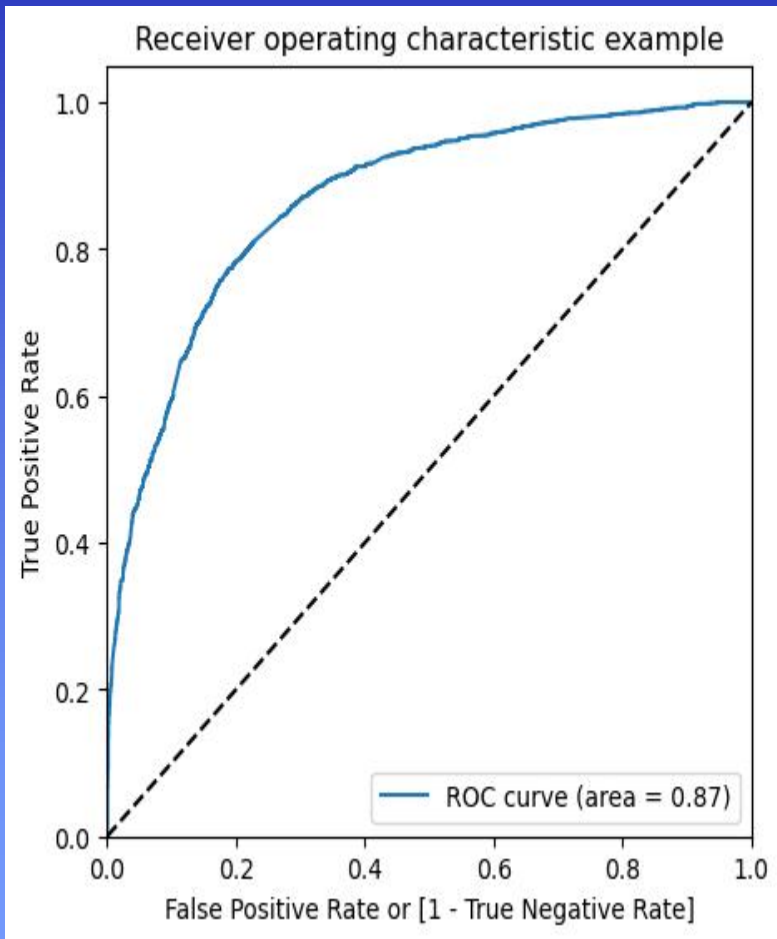
1. Total Visits
2. Total Time Spent On Website
3. Page Views Per Visit



CONFUSION METRIX ON TRAIN DATA



ROC CURVE and PROBABILITY METRIC PLOT



METRICS AND CUT OFF

- Optimal Probability Cutoff: 0.44

Evaluation Metrics:

- Accuracy :
 - Training Data - 79%
 - Test Data - 79.7%
- Sensitivity :
 - Training Data - 78.7%
 - Test Data - 78.3%
- Specificity :
 - Training Data - 79.4%
 - Test Data - 81.2%



KEY INSIGHTS

1. **Visits Matter:** More website visits increase the likelihood of conversion.
2. **Time Spent On The Website:** The time on the website correlates to the Lead Conversions.
3. **Top Sources:** Focus on **Lead Add Forms** and **Welingak Website** for better engagement.

